13 The total surface area of a cube is 294 cm^2.

Work out the volume of the cube.

A cube has 6 equal square faces.

$$6 \overline{)\, 2^2 9^5 4} \quad 0\ 4\ 9$$

Area of one face = $294 \div 6 = 49\ cm^2$.

1 mark
Divide the surface area by 6

Edge length = $\sqrt{49} = 7$ cm.

1 mark
Square-root the face area

Volume = $7 \times 7 \times 7 = 49 \times 7 = 343\ cm^3$.

1 mark
Multiply all three edge lengths

1 mark
Correct volume

343 ..... cm^3

(Total for Question 13 is 4 marks)